

1 Fixed Length Subnet Mask (FLSM)

1.1 Introduction

Fixed Length Subnet Mask (FLSM) is a subnetting technique in which a network is divided into multiple **equal-sized subnets**. Every subnet uses the **same subnet mask**, which means each subnet has an identical number of IP addresses and supports the same number of hosts.

FLSM is commonly used in networks where all departments, branches, or network segments require approximately the same number of hosts. Since all subnets have a uniform size, network design and management become simple and systematic.

In FLSM, subnet bits are borrowed from the host portion of the original network address. The number of borrowed bits determines the total number of subnets, while the remaining host bits determine the number of hosts available in each subnet.

Although FLSM is easy to configure and administer, it may lead to inefficient utilization of IP addresses when different subnets require different numbers of hosts. For networks with varying host requirements, Variable Length Subnet Mask (VLSM) provides a more efficient solution.

1.2 Key Characteristics of FLSM

- Every subnet has the same subnet mask.
- Every subnet contains the same number of IP addresses.
- Every subnet supports the same number of usable hosts.
- Network design is simple and easy to manage.
- Suitable for networks with equal host requirements.
- Uses fixed-size subnet allocation.
- Easy to calculate network and broadcast addresses.
- May waste IP addresses if subnet sizes differ significantly.

1.3 Important Formulae

1. Number of Required Subnets

$$2^n \geq \text{Required Number of Subnets}$$

where n is the number of borrowed subnet bits.

2. New Prefix Length

$$\text{New Prefix} = \text{Original Prefix} + \text{Borrowed Bits}$$

3. Number of Host Bits

$$\text{Host Bits} = 32 - \text{New Prefix}$$

4. Total IP Addresses per Subnet

$$2^{\text{Host Bits}}$$

5. Number of Usable Hosts

$$2^{\text{Host Bits}} - 2$$

(Subtracting one Network Address and one Broadcast Address.)

6. Block Size

$$\text{Block Size} = 256 - \text{Interesting Octet of the Subnet Mask}$$

1.4 General Steps for Solving FLSM Problems

1. Determine the total number of required subnets.
2. Calculate the number of subnet bits to borrow.
3. Determine the new subnet prefix.
4. Calculate the corresponding subnet mask.
5. Determine the block size.
6. Calculate the total number of IP addresses per subnet.
7. Calculate the number of usable hosts per subnet.
8. Determine the network address of each subnet.
9. Determine the first usable host address.
10. Determine the last usable host address.
11. Determine the broadcast address of each subnet.
12. Verify that all required subnets have been allocated correctly.

1.5 Advantages of FLSM

- Easy to understand and implement.
- Simplifies subnet calculations.
- Uniform subnet structure.
- Easy network administration.
- Suitable for small and medium-sized networks.
- Reduces routing complexity.
- Ideal when every subnet has similar host requirements.

1.6 Disadvantages of FLSM

- Inefficient utilization of IP addresses.
- Wastes address space when host requirements differ.
- Cannot allocate different subnet sizes.
- Less scalable for growing networks.
- Not suitable for enterprise networks with varying departmental sizes.

1.7 Applications of FLSM

FLSM is commonly used in:

- Small office networks.
- Computer laboratories.
- Educational institutions.
- Branch offices with equal-sized departments.
- Training and academic subnetting exercises.

1.8 Important Examination Tips

- Always calculate the required number of subnet bits first.
- Do not forget to subtract two addresses when calculating usable hosts.
- Identify the **interesting octet** before calculating the block size.
- Ensure subnet ranges do not overlap.
- Verify the network address, broadcast address, and host range for every subnet.
- Present subnet calculations systematically using tables whenever possible.
- Double-check binary calculations and prefix lengths before finalizing the answer.

Example 1: FLSM Network Design

A software company has been assigned the IPv4 address block **192.168.20.0/24**. The company consists of **4 departments**, and each department requires an equal-sized subnet.

Determine the following:

1. Determine the subnet prefix and subnet mask.
2. Determine the block size of each subnet.
3. Determine the network address of each subnet.
4. Determine the first and last usable host addresses.
5. Determine the broadcast address of each subnet.
6. Determine the number of usable hosts per subnet.

Step 1: Determine the Number of Required Subnets

Number of departments

$$N = 4$$

Required number of subnet bits,

$$2^n \geq N$$

$$2^2 = 4$$

Therefore,

$$n = 2$$

Hence, borrow **2 bits** from the host portion.

Step 2: Determine the New Prefix

Original prefix

$$/24$$

Borrowed bits

$$2$$

New prefix

$$24 + 2 = 26$$

Therefore,

$$\boxed{/26}$$

Step 3: Determine the Subnet Mask

Prefix

$$/26$$

Binary representation

$$11111111.11111111.11111111.11000000$$

Decimal representation

$$\boxed{255.255.255.192}$$

Step 4: Determine the Block Size

The interesting octet is the fourth octet.

$$\text{Block Size} = 256 - 192$$

$$= 64$$

Therefore,

$$\boxed{\text{Block Size} = 64}$$

Subnet increment

$$0, 64, 128, 192$$

Step 5: Determine the Number of Hosts

Remaining host bits

$$32 - 26 = 6$$

Total addresses

$$2^6 = 64$$

Usable hosts

$$64 - 2 = 62$$

Therefore,

$$\boxed{62 \text{ usable hosts per subnet}}$$

Step 6: Calculate Each Subnet

Subnet	Network Address	Host Range	Broadcast Address	Usable Hosts
1	192.168.20.0/26	192.168.20.1 – 192.168.20.62	192.168.20.63	62
2	192.168.20.64/26	192.168.20.65 – 192.168.20.126	192.168.20.127	62
3	192.168.20.128/26	192.168.20.129 – 192.168.20.190	192.168.20.191	62
4	192.168.20.192/26	192.168.20.193 – 192.168.20.254	192.168.20.255	62

Final Answer

Original Network	192.168.20.0/24
Required Subnets	4
Borrowed Bits	2
New Prefix	/26
Subnet Mask	255.255.255.192
Block Size	64
Total Subnets	4
Addresses per Subnet	64
Usable Hosts per Subnet	62

Summary

The network **192.168.20.0/24** is divided equally into **4 subnets** using the FLSM technique. Each subnet has the prefix **/26** (Subnet Mask: **255.255.255.192**), contains **64 IP addresses**, supports **62 usable hosts**, and the subnets begin at increments of **64** in the fourth octet.

Example 2: FLSM Network Planning

A college receives the IPv4 address block **172.18.0.0/22**. The network administrator plans to divide the network equally among **8 laboratories** using the FLSM technique.

Calculate the following:

1. Determine the new subnet prefix.
2. Determine the subnet mask.
3. Determine the block size.
4. List the network address of each subnet.
5. List the broadcast address of each subnet.
6. Determine the valid host range for every subnet.
7. Determine the maximum number of usable hosts in each subnet.

Step 1: Determine the Number of Required Subnets

Number of laboratories

$$N = 8$$

Required subnet bits,

$$2^n \geq N$$

$$2^3 = 8$$

Therefore,

$$n = 3$$

Hence, borrow **3 bits** from the host portion.

Step 2: Determine the New Prefix

Original prefix

$$/22$$

Borrowed subnet bits

$$3$$

New prefix

$$22 + 3 = 25$$

Therefore,

$$\boxed{/25}$$

Step 3: Determine the Subnet Mask

Prefix

$$/25$$

Binary representation

11111111.11111111.11111111.10000000

Decimal representation

$$\boxed{255.255.255.128}$$

Step 4: Determine the Block Size

The interesting octet is the fourth octet.

$$\text{Block Size} = 256 - 128$$

$$= 128$$

Therefore,

$$\boxed{\text{Block Size} = 128}$$

Subnet increment

0, 128

Since the original network is /22, it spans four consecutive Class-C networks:

172.18.0.0

to

172.18.3.255

Each /24 network is divided into two /25 subnets, resulting in

$$4 \times 2 = 8$$

subnets.

Step 5: Determine the Number of Hosts

Remaining host bits

$$32 - 25 = 7$$

Total addresses per subnet

$$2^7 = 128$$

Usable hosts

$$128 - 2 = 126$$

Therefore,

126 usable hosts per subnet

Step 6: Calculate All Subnets

Subnet	Network Address	Host Range	Broadcast Address	Usable Hosts
1	172.18.0.0/25	172.18.0.1 – 172.18.0.126	172.18.0.127	126
2	172.18.0.128/25	172.18.0.129 – 172.18.0.254	172.18.0.255	126
3	172.18.1.0/25	172.18.1.1 – 172.18.1.126	172.18.1.127	126
4	172.18.1.128/25	172.18.1.129 – 172.18.1.254	172.18.1.255	126
5	172.18.2.0/25	172.18.2.1 – 172.18.2.126	172.18.2.127	126
6	172.18.2.128/25	172.18.2.129 – 172.18.2.254	172.18.2.255	126
7	172.18.3.0/25	172.18.3.1 – 172.18.3.126	172.18.3.127	126
8	172.18.3.128/25	172.18.3.129 – 172.18.3.254	172.18.3.255	126

Final Answer

Original Network	172.18.0.0/22
Required Subnets	8
Borrowed Bits	3
New Prefix	/25
Subnet Mask	255.255.255.128
Block Size	128
Total Subnets	8
Addresses per Subnet	128
Usable Hosts per Subnet	126

Summary

The address block **172.18.0.0/22** contains a total of

$$2^{32-22} = 1024$$

IP addresses. By borrowing **3 bits**, the network is divided equally into **8 FLSM subnets**. Each subnet has a prefix of **/25** with subnet mask **255.255.255.128**. Every subnet contains **128 IP addresses**, of which **126 are usable** for hosts.

Example 3: Enterprise FLSM Design

A multinational company has been allocated the IPv4 address block **10.50.0.0/20**. The company has **16 branch offices**, and each branch must receive an equal-sized subnet using the FLSM method.

Determine the following:

1. Determine the required subnet prefix.
2. Determine the corresponding subnet mask.
3. Determine the block size.
4. Determine the network address of all subnets.
5. Determine the broadcast address of each subnet.
6. Determine the usable host range for each subnet.
7. Calculate the total number of usable hosts supported by each subnet.

Step 1: Determine the Number of Required Subnets

Number of branch offices

$$N = 16$$

Required subnet bits,

$$2^n \geq N$$

$$2^4 = 16$$

Therefore,

$$n = 4$$

Hence, borrow **4 bits** from the host portion.

Step 2: Determine the New Prefix

Original prefix

$$/20$$

Borrowed subnet bits

$$4$$

New prefix

$$20 + 4 = 24$$

Therefore,

$$/24$$

Step 3: Determine the Subnet Mask

Prefix

$$/24$$

Binary representation

11111111.11111111.11111111.00000000

Decimal representation

$$255.255.255.0$$

Step 4: Determine the Block Size

The interesting octet is the third octet.

$$\text{Block Size} = 256 - 240$$

$$= 16$$

Since the borrowed bits extend into the third octet, each subnet spans one complete /24 network.

Therefore,

Block Size = 1 in the fourth octet (each subnet is a /24 network)

The original /20 network covers

10.50.0.0

to

10.50.15.255

Hence, sixteen /24 subnets are created.

Step 5: Determine the Number of Hosts

Remaining host bits

$$32 - 24 = 8$$

Total addresses

$$2^8 = 256$$

Usable hosts

$$256 - 2 = 254$$

Therefore,

254 usable hosts per subnet

Step 6: Calculate All Subnets

Subnet	Network Address	Host Range	Broadcast Address	Usable Hosts
1	10.50.0.0/24	10.50.0.1 – 10.50.0.254	10.50.0.255	254
2	10.50.1.0/24	10.50.1.1 – 10.50.1.254	10.50.1.255	254
3	10.50.2.0/24	10.50.2.1 – 10.50.2.254	10.50.2.255	254
4	10.50.3.0/24	10.50.3.1 – 10.50.3.254	10.50.3.255	254
5	10.50.4.0/24	10.50.4.1 – 10.50.4.254	10.50.4.255	254
6	10.50.5.0/24	10.50.5.1 – 10.50.5.254	10.50.5.255	254
7	10.50.6.0/24	10.50.6.1 – 10.50.6.254	10.50.6.255	254
8	10.50.7.0/24	10.50.7.1 – 10.50.7.254	10.50.7.255	254
9	10.50.8.0/24	10.50.8.1 – 10.50.8.254	10.50.8.255	254
10	10.50.9.0/24	10.50.9.1 – 10.50.9.254	10.50.9.255	254
11	10.50.10.0/24	10.50.10.1 – 10.50.10.254	10.50.10.255	254
12	10.50.11.0/24	10.50.11.1 – 10.50.11.254	10.50.11.255	254
13	10.50.12.0/24	10.50.12.1 – 10.50.12.254	10.50.12.255	254
14	10.50.13.0/24	10.50.13.1 – 10.50.13.254	10.50.13.255	254
15	10.50.14.0/24	10.50.14.1 – 10.50.14.254	10.50.14.255	254
16	10.50.15.0/24	10.50.15.1 – 10.50.15.254	10.50.15.255	254

Final Answer

Original Network	10.50.0.0/20
Required Subnets	16
Borrowed Bits	4
New Prefix	/24
Subnet Mask	255.255.255.0
Total Subnets	16
Addresses per Subnet	256
Usable Hosts per Subnet	254
Original Address Range	10.50.0.0 – 10.50.15.255

Summary

The address block **10.50.0.0/20** contains

$$2^{32-20} = 4096$$

IP addresses. By borrowing **4 bits** from the host portion, the network is divided into

$$2^4 = 16$$

equal-sized FLSM subnets.

Each subnet has:

$$\text{Prefix} = /24$$

$$\text{Subnet Mask} = 255.255.255.0$$

Each subnet contains

$$2^{32-24} = 256$$

IP addresses, providing

$$254$$

usable host addresses for each branch office.